Chapter 1 Matrices and System of Equations

Section 1.3 Matrix Arithmetic

Definition (Matrix, plural: matrices)
An array of mn numbers $a_{ij}, i = 1, \cdots, m; j = 1, \cdots, n$

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

is called an $m \times n$ matrix, denoted by $A = (a_{ij})$, where A has m rows and n columns, and a_{ij} is called the (i, j)-entry of A in ith row and the jth column.

Definition (Row vector, column vector) A $1 \times n$ matrix is a *row vector*. An $m \times 1$ matrix a *column vector*.

Definition (Equal matrices) Two $m \times n$ matrices $A = (a_{ij})$ and $B = (b_{ij})$ are said to be *equal* if $a_{ij} = b_{ij}$ for each i and j.

Remark Two equal matrices are of the same size.

Example $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \neq \begin{pmatrix} 1 & 2 \\ 4 & 5 \\ 3 & 6 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Example $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 2 & a \\ 4 & b & 6 \end{pmatrix}$ if and only if $a = 3$ and $b = 5$.

Definition (Augmented matrix) When an $m \times r$ matrix B attached to an $m \times n$ matrix A, the *augmented matrix* is an $m \times (n + r)$ matrix denoted by $(A|B)$, and

$$(A|B) = \left(\begin{array}{ccc|ccc} a_{11} & \cdots & a_{1n} & b_{11} & \cdots & b_{1r} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{mn} & \cdots & a_{mn} & b_{m1} & \cdots & b_{mr} \end{array}\right).$$

Example If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$ and $B = \begin{pmatrix} 10 \\ 20 \\ 30 \end{pmatrix}$ then $(A|A) = \left(\begin{array}{cc|cc} 1 & 2 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 5 & 6 & 5 & 6 \end{array}\right)$

and $(A|B) = \left(\begin{array}{cc|c} 1 & 2 & 10 \\ 3 & 4 & 20 \\ 5 & 6 & 30 \end{array}\right)$.

Definition (Square matrix) An $m \times n$ matrix A is called a *square matrix* of order n if $m = n$.

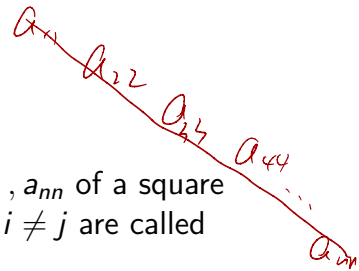
Example $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, B = \begin{pmatrix} 2 & 4 & 6 \\ 8 & 0 & 1 \\ 3 & 5 & 7 \end{pmatrix}$ are square matrices.

$C = \begin{pmatrix} 1 & 2 \end{pmatrix}, D = \begin{pmatrix} 2 & 4 \\ 8 & 0 \\ 3 & 5 \end{pmatrix}$ are no square matrices.

Definition (Square matrix) An $m \times n$ matrix A is called a *square matrix* of order n if $m = n$.

Example $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, B = \begin{pmatrix} 2 & 4 & 6 \\ 8 & 0 & 1 \\ 3 & 5 & 7 \end{pmatrix}$ are square matrices.

$C = \begin{pmatrix} 1 & 2 \end{pmatrix}, D = \begin{pmatrix} 2 & 4 \\ 8 & 0 \\ 3 & 5 \end{pmatrix}$ are no square matrices.



Definition (Diagonal / Off-diagonal) The entries $a_{11}, \cdots, a_{nn}$ of a square matrix are called *diagonal elements* while the entries $a_{ij}, i \neq j$ are called *off-diagonal elements*.

Example In the above example, the diagonal elements of A are $1, 4$. The diagonal elements of B are $2, 0, 7$.

Definition (Matrix Addition) If $A = (a_{ij})$ and $B = (b_{ij})$ are both $m \times n$ matrices, then *sum* $A + B$ is the $m \times n$ matrix whose (i, j) entry is $a_{ij} + b_{ij}$, i.e. $A + B = (a_{ij} + b_{ij})$.

Example Let

$$A = \begin{pmatrix} 1 & -1 & 4 \\ 2 & 3 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & -4 \\ 8 & -1 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 0 & 0 \\ 9 & -1/2 \end{pmatrix}$$

Then

$$A + B = \begin{pmatrix} 1+1 & (-1)+0 & 4+(-4) \\ 2+8 & 3+(-1) & 0+1 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 0 \\ 10 & 2 & 1 \end{pmatrix}$$

Since the sizes don't match, none of these is defined: $A + C$, $C + A$, $B + C$, $C + B$.

数乘矩阵

Definition (Scalar Multiplication (Scalar product)) If $A = (a_{ij})$ is an $m \times n$ matrix and α is a scalar, then *scalar multiple* αA *of* A *by* α is the $m \times n$ matrix whose (i, j) entry is αa_{ij}.

Example $5 \begin{pmatrix} 3 & -1 \\ -2 & 5 \\ 0 & -9 \end{pmatrix} = \begin{pmatrix} 15 & -5 \\ -10 & 25 \\ 0 & -45 \end{pmatrix}$.

scalar product

Definition (Matrix Multiplication) If $A = (a_{ij})$ is an $m \times n$ matrix and $B = (b_{ij})$ is an $n \times r$ matrix, then the *product* $AB = C = (c_{ij})$ is the $m \times r$ matrix whose entries are defined by

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} = a_{i1} b_{1j} + a_{i2} b_{2j} + \cdots + a_{in} b_{nj}.$$

$$AB = \begin{pmatrix} \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & & \vdots \end{pmatrix} \begin{pmatrix} & b_{1j} & \\ \cdots & b_{2j} & \cdots \\ & \vdots & \\ & b_{nj} & \end{pmatrix} = \begin{pmatrix} \vdots & \\ \cdots & c_{ij} \quad \cdots \\ \vdots & \end{pmatrix}.$$

Example Let $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} -2 & 0 & 1 \\ 3 & 1 & 1 \end{pmatrix}$.

$$AB = \begin{pmatrix} 1(-2) + 2(3) & 1(0) + 2(1) & 1(1) + 2(1) \\ 0(-2) + (-1)(3) & 0(0) + (-1)(1) & 0(1) + (-1)(1) \end{pmatrix} = \begin{pmatrix} 4 & 2 & 3 \\ -3 & -1 & -1 \end{pmatrix}.$$

Generally $AB \neq BA$, even the matrix multiplication is valid.

Example Compute $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 3 & 4 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

size didn't match

Generally $AB \neq BA$, even the matrix multiplication is valid.

Example Compute $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 3 & 4 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

Solution

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} (1)(0)+(0)(3) & (1)(0)+(0)(4) \\ (0)(0)+(0)(3) & (0)(0)+(0)(4) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

and

$$\begin{pmatrix} 0 & 0 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} (0)(1)+(0)(0) & (0)(0)+(0)(0) \\ (3)(1)+(4)(0) & (3)(0)+(4)(0) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 3 & 0 \end{pmatrix}.$$

If $AB = AC$ for a nonzero matrix A, then it is not true in general that $B = C$.

For example, if we consider

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix}, B = \begin{pmatrix} 3 & 4 \\ 0 & 0 \end{pmatrix}, C = \begin{pmatrix} 5 & 6 \\ 0 & 0 \end{pmatrix},$$

then

$$AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = AC$$

However $B \neq C$.

row and column exchange

Definition (Transpose) The transpose of an $m \times n$ matrix $A = (a_{ij})$ is the $n \times m$ matrix $B = (b_{ij})$ defined by $b_{ji} = a_{ij}$, for $i = 1, \cdots, m$ and $j = 1, \cdots, n$. The transpose of A is denoted by A^T.

In some books, the transpose of A is denoted by A^t or A'.

Example

$$\begin{pmatrix} 1 & 5 & 3 \\ 7 & 1 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 7 \\ 5 & 1 \\ 3 & 6 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -5 & -2 \\ 3 & 0 \end{pmatrix}^T = \begin{pmatrix} -5 & 3 \\ -2 & 0 \end{pmatrix}.$$

Example Let $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} -2 & 0 & 1 \\ 3 & 1 & 1 \end{pmatrix}$. Then

If $A \cdot B$ defined

$$B^T A^T = \begin{pmatrix} -2 & 3 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$$

$B \cdot A$ undefined

$$= \begin{pmatrix} 1(-2)+2(3) & 0(-2)+(-1)(3) \\ 1(0)+2(1) & 0(0)+(-1)(1) \\ 1(1)+2(1) & 0(1)+(-1)(1) \end{pmatrix} = \begin{pmatrix} 4 & -3 \\ 2 & -1 \\ 3 & -1 \end{pmatrix}.$$

Then $B^T \cdot A^T$ is defined

$$AB = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -2 & 0 & 1 \\ 3 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1(-2)+2(3) & 1(0)+2(1) & 1(1)+2(1) \\ 0(-2)+(-1)(3) & 0(0)+(-1)(1) & 0(1)+(-1)(1) \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 2 & 3 \\ -3 & -1 & -1 \end{pmatrix}.$$

Theorem

Each of the following statements is valid for any scalars α and for any matrices A, B for which the indicated operations are defined.

$$(A^T)^T = A$$
$$(\alpha A)^T = \alpha A^T$$
$$(A+B)^T = A^T + B^T$$
$$(AB)^T = B^T A^T$$

proof: $C = AB$

$A: m \times n$ $B: n \times p$

$C^T = (AB)^T$

$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$ $\quad (C^T)_{ij} = c_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki}$

$D = B^T \cdot A^T$

$d_{ij} = \sum_{k=1}^{n} (b^T)_{ik} (a^T)_{kj}$

$= \sum_{k=1}^{n} b_{ki} a_{jk}$

//

$$c_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki}$$

对称

Definition (Symmetric) A square matrix $A = (a_{ij})$ is called a symmetric matrix if $A = A^T$.

Example

$\begin{pmatrix} 1 & 5 & 3 \\ 5 & 2 & 6 \\ 3 & 6 & 0 \end{pmatrix}$ is symmetric. $\begin{pmatrix} 1 & 5 & 3 \\ 5 & 2 & 6 \\ 0 & 6 & 0 \end{pmatrix}$ is not symmetric.

对角 $\Rightarrow$ 对称

Definition (Symmetric) A square matrix $A = (a_{ij})$ is called a symmetric matrix if $A = A^T$.

$A \cdot A^T$ is always defined $\Rightarrow$ square

Example Let A be a matrix and let $B = AA^T$. Show that B is symmetric.

show $B^T = B$ [$(A \cdot B)^T = A^T \cdot B^T$]

Definition (Symmetric) A square matrix $A = (a_{ij})$ is called a symmetric matrix if $A = A^T$.

Example Let A be a matrix and let $B = AA^T$. Show that B is symmetric.

Solution [Idea: We have to show $B^T = B$.]

$$\begin{aligned} B^T &= (AA^T)^T \\ &= (A^T)^T A^T \\ &= AA^T \\ &= B \end{aligned}$$

Since $B^T = B$, B is symmetric.

Chapter 1 Matrices and System of Equations

Section 1.4 Matrix Algebra

Theorem (Addition) For any $m \times n$ matrices A, B, and C, the following statements are true.

$$\begin{aligned} A + B &= B + A \\ (A + B) + C &= A + (B + C) \end{aligned}$$

Definition (Zero matrix) An $m \times n$ matrix whose entries are all 0 is called a *zero matrix*, denoted by $O_{m,n}$, $O_{m\times n}$ or simply O.
If $\mathbf{a}$ is a column vector with all entries to be zero, it is called a zero column vector, denoted by $\mathbf{0}$.

The zero matrix is the identity element for matrix addition. For example,

$$\begin{pmatrix} 3 & 1 & 2 \\ 5 & 0 & 9 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 1 & 2 \\ 5 & 0 & 9 \end{pmatrix}$$

Definition (Additive inverse) If $A = (a_{ij})$ is an $m \times n$ matrix, then the *additive inverse* of A is $(-1)A$.

Example The additive inverse of $\begin{pmatrix} 3 & 1 \\ 2 & 5 \\ 0 & 9 \end{pmatrix}$ is $\begin{pmatrix} -3 & -1 \\ -2 & -5 \\ -0 & -9 \end{pmatrix}$, because

$$\begin{pmatrix} 3 & 1 \\ 2 & 5 \\ 0 & 9 \end{pmatrix} + \begin{pmatrix} -3 & -1 \\ -2 & -5 \\ -0 & -9 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Notation Denote the additive inverse of A by $-A$. If B is a matrix, denote $B + (-A)$ by $B - A$.

Here are some properties for any $m \times n$ matrices A, B and C.

	Addition
Closure	$A + B$ is an $m \times n$ matrix
Associativity	$A + (B + C) = (A + B) + C$
Commutativity	$A + B = B + A$
Existence of an identity element	There is O such that $A + O = A$.
Existence of inverse elements	$A + (-A) = O$

Same size

相反数

scalar can be changed in any position

Theorem (Scalar Multiplication) Each of the following statements are true for any scalars α and β and for any matrices A, B for which *the indicated operations are defined*.

$$\begin{aligned}
(\alpha\beta)A &= \alpha(\beta A) \\
\alpha(A+B) &= \alpha A + \alpha B \\
(\alpha+\beta)A &= \alpha A + \beta A \\
\alpha(AB) &= (\alpha A)B = A(\alpha B)
\end{aligned}$$

Theorem (Multiplication and distributive law) Each of the following statements are true for any matrices A, B, and C for which *the indicated operations are defined*.

$$\begin{aligned} AB &\neq BA, \quad \text{in general} \\ (AB)C &= A(BC) \\ A(B+C) &= AB+AC \\ (A+B)C &= AC+BC \end{aligned}$$

Example $(A+B)(A-B) = A(A-B)+B(A-B) = AA-AB+BA-BB = A^2-AB+BA-B^2$, which might not be A^2-B^2.

Definition (Power) For a square matrix A, A^k is the product of k A's, i.e. $A^k = AAAA\cdots A$ (k copies).

$\sim$: tilde $\quad A^T = (\tilde{a}_{ij}) \quad a_{ji} = \tilde{a}_{ij}$

Definition (Identity (Unit) matrix) An $n \times n$ matrix $A = (a_{ij})$ is called the *identity matrix* of order n if

$$a_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

and denoted by $A = I_n$, or simply $A = I$.

unit matrix

Example $I_1 = (1)$, $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, etc.

Theorem For any $m \times n$ matrix A, $AI_n = A$ and $I_m A = A$.

Here are some properties for any $n \times n$ matrices A, B and C

	Addition	Multiplication
Closure	$A + B$ is an $n \times n$ matrix	AB is an $n \times n$ matrix
Associativity	$A+(B+C)=(A+B)+C$	$A(BC) = (AB)C$
Commutativity	$A + B = B + A$	No!
Existence of an identity element	$A + O = A$	$AI = A$ $IA = A$
Existence of inverse elements	$A + (-A) = O$	?
Distributivity	$A(B + C) = AB + AC$ $(A + B)C = AC + BC$	

非奇异矩阵

Definition (Nonsingular matrix / Invertible matrix) A square matrix A of order n is said to be *nonsingular/invertible* if there exists a matrix B such that $AB = BA = I_n$. The matrix B is said to be a *multiplicative inverse* of A, and denote $B = A^{-1}$.

逆阵

Uniqueness of inverse If matrices B and C are (multiplicative) inverses of a matrix A, then $B = C$.

only one inverse

Proof If B and C are both multiplicative inverses of A, then

$$B = BI = B(AC) = (BA)C = IC = C.$$

反证法

Thus, a matrix can have at most one multiplicative inverse.

Example Are the matrices $A = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix}$ multiplicative inverse of each other?

Example Are the matrices $A = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix}$ multiplicative inverse of each other?

Solution Since $AB = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $BA = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, A and B are multiplicative inverse of each other.

It is not necessary that a square matrix has its inverse.

Example Show that $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ has no inverse.

Solution Suppose A has an inverse $B = (b_{ij})$. Then

$$I = BA = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} b_{11} & 0 \\ b_{21} & 0 \end{pmatrix} \neq \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

It is contradictory to have $I \neq I$. Hence the assumption that A has an inverse is false.

Definition (Singular) A square matrix is said to be *singular* if it does not have a multiplicative inverse.

Mathematical Induction 归纳法

Statement $P(n)$, $n \in \mathbb{N}$

① Show $P(1)$ is true

★ ② assume $P(k)$ is true for a $k \in \mathbb{N}$ $\Rightarrow$ show $P(k+1)$ is true

eg. prove $2^n > n^2$, $n \geq 5$

① $n=5$. $2^5 > 5^2$

② $n=k$. $k>5$. $2^k = \left(2^{\frac{k}{2}}\right)^2$ $\quad k^2$

$2^{\frac{k}{2}} > 2^{\frac{5}{2}} >$

③ $P(k+1) = 2^{k+1} = 2 \cdot 2^k > 2 \cdot k^2 > (k+1)^2$

$\Rightarrow 2^{k+1} > (k+1)^2$

$\Rightarrow P(k+1)$ is true

$2k^2 \geq (k+1)^2$

$= k^2 - 2k - 1$, $k > 5$

$\frac{2 \pm \sqrt{8}}{2} = 1 \pm \sqrt{2} < 5$

So $k^2 - 2k - 1 > 0$.

$P(n)$: $(A_1 A_2 \cdots A_n)^{-1} = A_n^{-1} A_{n-1}^{-1} \cdots A_1^{-1}$ $\quad n = 1, 2, 3 \cdots$

① $P(1) = A_1^{-1} = A_1^{-1}$

② assume $P(k)$ is true

$P(k) = (A_1 \cdots A_k)^{-1} = A_k^{-1} \cdots A_1^{-1}$

③ show $P(k+1)$ is true

$P(k+1) = (A_1 \cdots A_{k+1})^{-1} = A_{k+1}^{-1} A_k^{-1} \cdots A_1^{-1} = \left[(A_1 \cdots A_k)(A_{k+1})\right]^{-1}$

$= A_{k+1}^{-1} \cdot (A_k^{-1} \cdots A_1^{-1})$

$P(k+1) = (A_1 \cdots A_{k+1})^{-1} = A_{k+1}^{-1} \cdots A_1^{-1}$

$P(k+1)$ is true

Theorem If A and B are nonsingular matrices, then AB is also nonsingular and $(AB)^{-1} = B^{-1}A^{-1}$.

Proof

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I$$

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$$

Hence, AB is nonsingular and $(AB)^{-1} = B^{-1}A^{-1}$.

Exercise Prove by the induction that, if $A_1, \cdots, A_n$ are all nonsingular with the same size, then the product $A_1A_2 \cdots A_n$ is nonsingular and

$$(A_1A_2 \cdots A_n)^{-1} = A_n^{-1} \cdots A_2^{-1}A_1^{-1}.$$

(This n is not the size of the matrix.)

Theorem If A is a nonsingular matrix and $r \neq 0$ is a scalar, then rA and A^T are nonsingular. Furthermore,

- $(rA)^{-1} = \frac{1}{r}A^{-1}$;
- $(A^T)^{-1} = (A^{-1})^T$.

Proof Exercise.

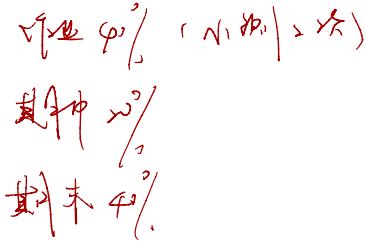
$$(A^T)^{-1} = (A^{-1})^T$$

$$(A^T)^{-1} \cdot A^T = I = I^T = (A \cdot A^{-1})^T$$

$$= (A^{-1})^T \cdot A^T$$

Chapter 1 Matrices and System of Equations

Section 1.1 Systems of Linear Equations



Definition (Linear equation) A *linear equation* is an equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where $a_1, \cdots, a_n$ and b are all known in advance.

$x_1, \cdots, x_n$ are called the *variables* in the above case.

We sometimes write x, y, z, w, etc. for variables instead of $x_1, \cdots, x_n$.

Example $4x_1 - 5x_2 + 2 = x_1$ and $x_2 = 2(\sqrt{6} - x_1) + x_3$ are linear equations.

$4x_1 - 5x_2 + 2 = x_1$ and $x_2 = 2(\sqrt{6} - x_1) + x_3$

↓ ↓

rearranged rearranged

↓ ↓

$3x_1 - 5x_2 = -2$ $2x_1 + x_2 - x_3 = 2\sqrt{6}$

Definition (Linear System) A *linear system of m equations in n unknowns* is a system of the form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m \end{cases}$$

Example $\begin{cases} x_1 + 2x_2 + x_4 = 7 \\ x_1 + x_2 + x_3 - x_4 = 3 \end{cases}$

Write a linear system using matrix multiplication

The linear system $\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ \vdots & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m \end{cases}$ can be expressed as

$$A\mathbf{x} = \mathbf{b}$$

where A is the **Coefficient Matrix**,

column

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}.$$

Definition (Solution) A *solution* to the system of m equations in n unknowns $x_1, x_2, \cdots, x_n$ is a list of numbers

$$(x_1, x_2, \cdots, x_n) = (s_1, s_2, \cdots, s_n)$$

that satisfy *all* m equations.

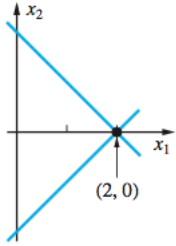
Consistency of a system of equations

If a system has at least one solution, we say that it is consistent.
If a system has no solution, we say that the system is inconsistent.

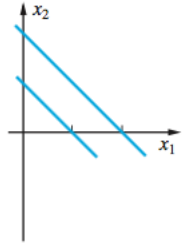
(i) $x_1 + x_2 = 2$
$x_1 - x_2 = 2$

(ii) $x_1 + x_2 = 2$
$x_1 + x_2 = 1$

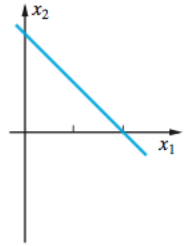
(iii) $x_1 + x_2 = 2$
$-x_1 - x_2 = -2$



(i)



(ii)

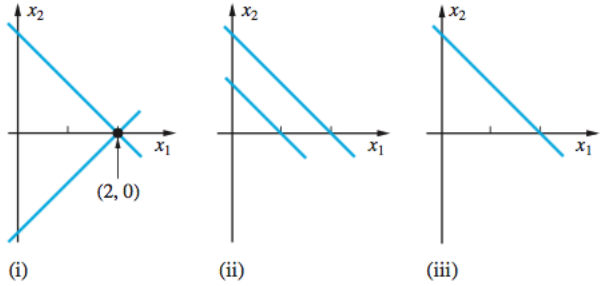


(iii)

(i) $x_1 + x_2 = 2$
$x_1 - x_2 = 2$

(ii) $x_1 + x_2 = 2$
$x_1 + x_2 = 1$

(iii) $x_1 + x_2 = 2$
$-x_1 - x_2 = -2$



Basic Facts: A system of linear equations has either

(i) **unique solution** (consistent) or

(ii) **no solution** (inconsistent) or

(iii) **infinitely many solutions** (consistent) .

Definition (Equivalent Systems) Two systems of equations involving the same variables are said to be *equivalent* if they have the same set of solution.

same solution

Three elementary operations that give equivalent systems

I. The order in which any two equations are written may be interchanged.

Notation: $R_i \leftrightarrow R_j$.

Example Interchange two equations ($R_1 \leftrightarrow R_2$).

The systems $\begin{cases} 2x_1 + x_2 = 8 \\ x_1 + x_2 = 3 \end{cases}$ and $\begin{cases} x_1 + x_2 = 3 \\ 2x_1 + x_2 = 8 \end{cases}$ are equivalent.

II. Both sides of an equation may be multiplied by the same nonzero real number c. Notation: $cR_i \to R_i$.

Example Multiply the second equation by 3 ($3R_2 \to R_2$).

The systems $\begin{cases} 2x_1 + x_2 = 8 \\ x_1 + x_2 = 3 \end{cases}$ and $\begin{cases} 2x_1 + x_2 = 8 \\ 3x_1 + 3x_2 = 9 \end{cases}$ are equivalent.

III. A multiple of the ith equation is added to j^{th} equation. Notation: $cR_i + R_j \to R_j$.

Example $-1/2$ times the first equation, added to the second equation $-1/2R_1 + R_2 \to R_2$.

The systems $\begin{cases} 2x_1 + x_2 = 8 \\ x_1 + x_2 = 3 \end{cases}$ and $\begin{cases} 2x_1 + x_2 = 8 \\ 1/2x_2 = -1 \end{cases}$ are equivalent.

Definition (Strict triangular form) A system is said to be in *strict triangular form* if, in the kth equation, the coefficients of the first $k-1$ variables are all zero and the coefficient of x_k is nonzero ($k = 1, \cdots, n$).

Example The system of equations

$$\begin{cases} 3x_1 + & 2x_2 - & x_3 = & 1 \\ 0x_1 + & 1x_2 + & 0x_3 = & 3 \\ 0x_1 + & 0x_2 + & 2x_3 = & 4 \end{cases}$$

is of strict triangular form.

Example Solve $\begin{cases} x_1 + 2x_2 + x_3 &= 3 \\ 3x_1 - x_2 - 3x_3 &= -1 \\ 2x_1 + 3x_2 + x_3 &= 4 \end{cases}$

Outline of solution

Stage 1 Reduce the system to a strict/upper triangular form

Perform elementary operations to give an equivalent system

$$\begin{cases} x_1 + 2x_2 + x_3 &= 3 \\ -7x_2 - 6x_3 &= -10 \\ (-1/7)x_3 &= (-4/7) \end{cases}$$

Stage 2 Backward substitution method

In the last equation of that system, x_1 and x_2 are eliminated. Then we solve for x_3 and then substitute the value of x_3 to find x_2, and then x_1.

Stage 1 Perform elementary operations to change

$$\begin{cases} x_1 + 2x_2 + x_3 = 3 \\ 3x_1 - x_2 - 3x_3 = -1 \\ 2x_1 + 3x_2 + x_3 = 4 \end{cases} \text{ to } \begin{cases} x_1 + 2x_2 + x_3 = 3 \\ -7x_2 - 6x_3 = -10 \\ (-1/7)x_3 = (-4/7) \end{cases}$$

Detail: Eliminate x_1 in Row 2 by replacing Row 2 with "-3(Row 1)+Row 2"

$$\begin{cases} x_1 + 2x_2 + x_3 = 3 \\ 0x_1 - 7x_2 - 6x_3 = -10 \\ 2x_1 + 3x_2 + x_3 = 4 \end{cases}$$

Eliminate x_1 in Row 3 by replacing Row 3 with "-2(Row 1)+Row 3"

$$\begin{cases} x_1 + 2x_2 + x_3 = 3 \\ -7x_2 - 6x_3 = -10 \\ 0x_1 - x_2 - x_3 = -2 \end{cases}$$

Eliminate x_2 in Row 3 by replacing Row 3 with "$\frac{-1}{7}$(Row 2)+Row 3"

$$\begin{cases} x_1 + 2x_2 + x_3 = 3 \\ -7x_2 - 6x_3 = -10 \\ 0x_2 + (-1/7)x_3 = (-4/7) \end{cases}$$

Stage 2 Solve x_1, x_2, x_3 from $\begin{cases} x_1 + 2x_2 & +x_3 & = 3 \\ -7x_2 & -6x_3 & = -10 \\ & -\frac{1}{7}x_3 & = -\frac{4}{7} \end{cases}$ by back substitution.

Detail From Row (3),

$$x_3 = 4.$$

Substitute $x_3 = 4$ into Row (2),

$$7x_2 = -10 + 6(4)$$
$$\text{and so} \quad x_2 = -2.$$

Substitute $x_3 = 4$ and $x_2 = -2$ into Row (1),

$$x_1 = 3 - 4 - 2(-2) = 3.$$

So, the solution is $(x_1, x_2, x_3) = (4, -2, 3)$.

Matrix Representation of a Linear System

$$\begin{cases} a_{11}x_1 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + \ldots + a_{2n}x_n &= b_2 \\ & \vdots \\ a_{m1}x_1 + \ldots + a_{mn}x_n &= b_m \end{cases} \quad \rightarrow \quad A\mathbf{x} = \mathbf{b}$$

Define $[A|\mathbf{b}]$ as the $m \times (n+1)$ augmented matrix

$$[A|\mathbf{b}] = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \ldots & a_{1n} & b_1 \\ a_{21} & a_{22} & \ldots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} & b_m \end{array}\right)$$

Example The system of linear equations

$$\begin{aligned} 2x_1 + 4x_2 - 3x_3 + 5x_4 + x_5 &= 9 \\ 3x_1 + x_2 + \quad x_4 - 3x_5 &= 0 \\ -2x_1 + 7x_2 - 5x_3 + 2x_4 + 2x_5 &= -3 \end{aligned}$$

has matrix representation $A\mathbf{x} = \mathbf{b}$ with the augmented matrix $[A|\mathbf{b}]$ as

$$\left(\begin{array}{ccccc|c} 2 & 4 & -3 & 5 & 1 & 9 \\ 3 & 1 & 0 & 1 & -3 & 0 \\ -2 & 7 & -5 & 2 & 2 & -3 \end{array}\right).$$

Elementary Row Operations

(I) (*Interchange*) Interchange two rows.

$$R_i \leftrightarrow R_j$$

(II) (*Scaling*) Multiply a row by a nonzero constant.

$$cR_i \to R_i, \quad c \neq 0$$

(III) (*Replacement*) Replace a row by its sum with a multiple of another row.

$$R_i + cR_j \to R_i$$

Elementary row operations Corresponding to each type of operations on equations, we can perform the following operations on the augmented matrix.

I. Interchange two rows.

Example

$$\left(\begin{array}{cc|c} 2 & 1 & 8 \\ 1 & 1 & 3 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 1 & 8 \end{array}\right)$$

II. Multiply a row by a nonzero real number.

Example

$$\left(\begin{array}{cc|c} 2 & 1 & 8 \\ 1 & 1 & 3 \end{array}\right) \xrightarrow{3R_2 \to R_2} \left(\begin{array}{cc|c} 2 & 1 & 8 \\ 3 & 3 & 9 \end{array}\right)$$

III. Replace a row by its sum with a multiple of another row.

Example

$$\left(\begin{array}{cc|c} 2 & 1 & 8 \\ 1 & 1 & 3 \end{array}\right) \xrightarrow{-1/2R_1 + R_2 \to R_2} \left(\begin{array}{cc|c} 2 & 1 & 8 \\ 0 & 1/2 & -1 \end{array}\right)$$

Example Solve $\begin{cases} 2x - 3y - z + 2w = -2 \\ x + 0y + 3z + 1w = 6 \\ 2x - 3y - z + 3w = -3 \\ 0x + y + z - 2w = 4 \end{cases}$

Solution

$$\left[\begin{array}{cccc|c} 2 & -3 & -1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 6 \\ 2 & -3 & -1 & 3 & -3 \\ 0 & 1 & 1 & -2 & 4 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 2 & & & & \\ 0 & & & & \\ 0 & & & & \\ 0 & & & & \end{array}\right]$$

Then we apply back substitution method.
By Row 4, $w = -1$.
Substitute $w = -1$ into Row 3, $z = 2$.
Substitute $w = -1$ and $z = 2$ into Row 2, $y = 0$.
Substitute $w = -1$, $z = 2$ and $y = 0$ into Row 1, $x = 1$.
The solution is $(1, 0, 2, -1)$.

Exercise Solve $\begin{cases} x_1 + 2x_2 + 2x_3 &= 4 \\ x_1 + 3x_2 + 3x_3 &= 5 \\ 2x_1 + 6x_2 + 5x_3 &= 6 \end{cases}$.

Solution

$$\left(\begin{array}{ccc|c} 1 & 2 & 2 & 4 \\ 1 & 3 & 3 & 5 \\ 2 & 6 & 5 & 6 \end{array}\right) \xrightarrow{-1R_1+R_2\to R_2} \left(\begin{array}{ccc|c} 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 \\ 2 & 6 & 5 & 6 \end{array}\right) \xrightarrow{-2R_1+R_3\to R_3}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 2 & 1 & -2 \end{array}\right) \xrightarrow{-2R_2+R_3\to R_3} \left(\begin{array}{ccc|c} 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -1 & -4 \end{array}\right)$$

Solve x_3 from Row 3, $x_3 = 4$.
Substitute $x_3 = 4$ back into Row 2, $x_2 = -3$.
Substitute $x_3 = 4$ and $x_2 = -3$ into Row 1, $x_1 = 2$.
So $(x_1, x_2, x_3) = (2, -3, 4)$ is the solution.

Chapter 1 Matrices and System of Equations

Section 1.2 Row Echelon Form

阶梯形，梯队

/ˈe ʃə lɑn/

Definition (Row Echelon Form) A matrix is said to be in *row echelon form* if

(1) the first nonzero entry in each nonzero row is 1;

階形

leading 1

$$\begin{pmatrix} \boxed{1} & \star & \star & \star & \star & \star & \star & \cdots & \star & \star \\ 0 & 0 & \boxed{1} & \star & \star & \star & \star & \cdots & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & \boxed{1} & \cdots & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & \boxed{1} & \star \\ & & & & \cdots & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}$$

where $\star$ could be any number

Definition (Row Echelon Form) A matrix is said to be in *row echelon form* if

(ii) if row k does not consist entirely of zeros, the number of leading zero entries in row $k+1$ is greater than the number of leading zero entries in row k;

$$\left(\begin{array}{cccc|cccccc|c} 1 & \star & \star & \star & \star & \star & \star & \cdots & \star & \star \\ 0 & 0 & 1 & \star & \star & \star & \star & \cdots & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & \cdots & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 1 & \star \\ & & & & \cdots & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right)$$

where $\star$ could be any number

Definition (Row Echelon Form) A matrix is said to be in *row echelon form* if

(iii) if there are rows whose entries are all zero, they are below the rows having nonzero entries.

$$\left(\begin{array}{cccccccccc|c}
1 & \star & \star & \star & \star & \star & \star & \cdots & \star & \star \\
0 & 0 & 1 & \star & \star & \star & \star & \cdots & \star & \star \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & \cdots & \star & \star \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 1 & \star \\
 & & & & \cdots & & & & & \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0
\end{array}\right)$$

where $\star$ could be any number

have to be on the bottom.

Definition (Row Echelon Form) A matrix is said to be in *row echelon form* if

(i) the first nonzero entry in each nonzero row is 1;

(ii) if row k does not consist entirely of zeros, the number of leading zero entries in row $k+1$ is greater than the number of leading zero entries in row k;

(iii) if there are rows whose entries are all zero, they are below the rows having nonzero entries.

$$\left(\begin{array}{cccc|cccccc} 1 & \star & \star & \star & \star & \star & \star & \cdots & \star & \star \\ 0 & 0 & 1 & \star & \star & \star & \star & \cdots & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & \cdots & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 1 & \star \\ & & & & \cdots & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right)$$

where $\star$ could be any number

Example $\begin{pmatrix} 1 & 4 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & 3 & 1 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ are in row echelon form (ref).

$$\left(\begin{array}{ccccccccc|c} 1 & \star & \star & \star & \star & \star & \star & \cdots & \star & \star \\ 0 & 0 & 1 & \star & \star & \star & \star & \cdots & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & \cdots & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 1 \\ & & & & \cdots & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \end{array}\right)$$

where $\star$ could be any number

Extra Exercises

Determine the following matrices are in row echelon form or not:

$$\left(\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & -1 \end{array}\right), \left(\begin{array}{ccc|c} 2 & 4 & 6 & 3 \\ 0 & 3 & 5 & 1 \\ 0 & 0 & 4 & 2 \end{array}\right), \left(\begin{array}{ccc|c} 0 & 0 & 0 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right),$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \left(\begin{array}{cccc|c} 1 & 4 & 2 & 1 & 3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{array}\right), \left(\begin{array}{cc|c} 0 & 1 & 0 \\ 1 & 0 & 1 \end{array}\right)$$

Extra Exercises

Determine the following matrices are in row echelon form or not:

$$\left(\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & -1 \end{array}\right), \left(\begin{array}{ccc|c} 2 & 4 & 6 & 3 \\ 0 & 3 & 5 & 1 \\ 0 & 0 & 4 & 2 \end{array}\right), \left(\begin{array}{ccc|c} 0 & 0 & 0 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right),$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 7 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right), \left(\begin{array}{cccc|c} 1 & 4 & 2 & 1 & 3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{array}\right), \left(\begin{array}{cc|c} 0 & 1 & 0 \\ 1 & 0 & 1 \end{array}\right)$$

Solution:

$$\begin{pmatrix} Y & N & N \\ Y & N & N \end{pmatrix}$$

/gɔ ʃɪən/

Gaussian Elimination The process of using elementary row operations I, II, and III to transform a linear system into one whose augmented matrix is in row echelon form is called *Gaussian Elimination*.

Example

Solve $\begin{cases} -x - y + 3z = 3 \\ x + z = 3 \\ 3x - y + 7z = 15 \end{cases}$ using Gaussian Elimination.

$$\left[\begin{array}{ccc|c} -1 & -1 & 3 & 3 \\ 1 & 0 & 1 & 3 \\ 3 & -1 & 7 & 15 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} -1 & -1 & 3 & 3 \\ 0 & -1 & 4 & 6 \\ 0 & -4 & 16 & 24 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} -1 & -1 & 3 & 3 \\ 0 & -1 & 4 & 6 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Gaussian Elimination The process of using elementary row operations I, II, and III to transform a linear system into one whose augmented matrix is in row echelon form is called *Gaussian Elimination*.

Example

Solve $\begin{cases} -x - y + 3z = 3 \\ x + z = 3 \\ 3x - y + 7z = 15 \end{cases}$ using Gaussian Elimination.

Solution

$$\left(\begin{array}{ccc|c} -1 & -1 & 3 & 3 \\ 1 & 0 & 1 & 3 \\ 3 & -1 & 7 & 15 \end{array}\right) \xrightarrow[3R_1+R_3\to R_3]{R_1+R_2\to R_2} \left(\begin{array}{ccc|c} -1 & -1 & 3 & 3 \\ 0 & -1 & 4 & 6 \\ 0 & -4 & 16 & 24 \end{array}\right)$$

$$\xrightarrow{-4R_2+R_3\to R_3} \left(\begin{array}{ccc|c} -1 & -1 & 3 & 3 \\ 0 & -1 & 4 & 6 \\ 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow[-R_1\to R_1]{-R_2\to R_2} \left(\begin{array}{ccc|c} 1 & 1 & -3 & -3 \\ 0 & 1 & -4 & -6 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 1 & -3 & -3 \\ 0 & 1 & -4 & -6 \\ 0 & 0 & 0 & 0 \end{array}\right) \qquad \left\{\begin{array}{rrcll} x & +y & -3z & = & -3 \quad (1) \\ & y & -4z & = & -6 \quad (2) \\ & & 0 & = & 0 \quad (3) \end{array}\right.$$

$$\left(\begin{array}{ccc|c} 1 & 1 & -3 & -3 \\ 0 & 1 & -4 & -6 \\ 0 & 0 & 0 & 0 \end{array}\right) \qquad \left\{\begin{array}{rrcll} x & +y & -3z & = & -3 \quad (1) \\ & y & -4z & = & -6 \quad (2) \\ & & 0 & = & 0 \quad (3) \end{array}\right.$$

(3) is always true.
From (2), $y = -6 + 4z$.
Substitute $y = -6 + 4z$ into (1), $x + (-6 + 4z) - 3z = -3$ and so $x = 3 - z$.

$$\left(\begin{array}{ccc|c} 1 & 1 & -3 & -3 \\ 0 & 1 & -4 & -6 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

z is free variable

$$\begin{cases} x + y - 3z = -3 & (1) \\ y - 4z = -6 & (2) \\ 0 = 0 & (3) \end{cases}$$

(3) is always true.
From (2), $y = -6 + 4z$.
Substitute $y = -6 + 4z$ into (1), $x + (-6 + 4z) - 3z = -3$ and so $x = 3 - z$.
So, the solution are of the form $(3 - \alpha, -6 + 4\alpha, \alpha)$, where α is any real number.

For example, when $\alpha = 1$, $(3 - \alpha, -6 + 4\alpha, \alpha) = (2, -2, 1)$ and $(2, -2, 1)$ is a solution of the system. When $\alpha = 2$, $(3 - \alpha, -6 + 4\alpha, \alpha) = (1, 2, 2)$ and $(1, 2, 2)$ is a solution of the system.

Definition (Leading variable) In each row of a row echelon form, the first variable with a nonzero coefficient is the rows *leading variable*.

Definition (Leading variable) In each row of a row echelon form, the first variable with a nonzero coefficient is the rows *leading variable*.

Or, in a row echelon form, the variable corresponding to *leading 1's* is the *leading variable*.

In the above example, x and y are leading variables.

Definition (Leading variable) In each row of a row echelon form, the first variable with a nonzero coefficient is the rows *leading variable*.

Or, in a row echelon form, the variable corresponding to *leading 1's* is the *leading variable*.

In the above example, x and y are leading variables.

Definition (Free variable) In a row echelon form, the variables that are not leading are *free variables*.

In the above example, z is the free variable.

Example

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 1 & 8 \\ 2 & 3 & 2 & 13 \end{array}\right) \xrightarrow[-2R_1+R_3\to R_3]{-R_1+R_2\to R_2} \xrightarrow{-R_2+R_3\to R_3} \xrightarrow{-R_3\to R_3} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{array}\right)$$

Last equation gives $0x_1 + 0x_2 + 0x_3 = 1$. Since it is always false that $0 = 1$, this system is inconsistent and has no solution.

Example

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 1 & 8 \\ 2 & 3 & 2 & 13 \end{array}\right) \xrightarrow[-2R_1+R_3\to R_3]{-R_1+R_2\to R_2} \xrightarrow{-R_2+R_3\to R_3} \xrightarrow{-R_3\to R_3} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{array}\right)$$

Last equation gives $0x_1 + 0x_2 + 0x_3 = 1$. Since it is always false that $0 = 1$, this system is inconsistent and has no solution.

Rule If the row echelon form of the augmented matrix contains a row of the form $\left(\begin{array}{ccc|c} 0 & 0 & \cdots & 0 & 1 \end{array}\right)$, then the system is inconsistent and has no solution.

Reduced Row Echelon Form A matrix is said to be in *reduced row echelon form* if

(i) the matrix is in row echelon form, and

(ii) the first nonzero entry in each row is the only nonzero entry in its column.

leading 1

stricker form

Example The following matrices are in reduced row echelon form.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 2 & 4 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Reduced Row Echelon form

Example The blue entries make the row echelon forms not reduced.

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 3 & 2 & 4 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

just Echelon Form

高斯消元法

Gauss-Jordan reduction The process of using elementary row operations to transform a matrix into reduced row echelon form is called *Gauss-Jordan reduction*.

Gauss-Jordan reduction The process of using elementary row operations to transform a matrix into reduced row echelon form is called *Gauss-Jordan reduction*.

Example Solve $\begin{cases} x_1 - x_2 - 2x_4 = 2 \\ x_1 + x_2 + 3x_3 + x_4 = 1 \\ - x_2 + x_3 - x_4 = 0 \end{cases}$ using Gauss-Jordan reduction.

Gauss-Jordan reduction The process of using elementary row operations to transform a matrix into reduced row echelon form is called *Gauss-Jordan reduction*.

Example Solve $\begin{cases} x_1 - x_2 - 2x_4 = 2 \\ x_1 + x_2 + 3x_3 + x_4 = 1 \\ - x_2 + x_3 - x_4 = 0 \end{cases}$ using Gauss-Jordan reduction.

Solution We first reduce the matrix to its row echelon form.

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 1 & 1 & 3 & 1 & 1 \\ 0 & -1 & 1 & -1 & 0 \end{array}\right) \xrightarrow{-1R_1+R_2\to R_2} \left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 0 & 2 & 3 & 3 & 1 \\ 0 & -1 & 1 & -1 & 0 \end{array}\right)$$

$$\xrightarrow{(1/2)R_2+R_3\to R_3} \left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 0 & 2 & 3 & 3 & 1 \\ 0 & 0 & \frac{5}{2} & \frac{1}{2} & -\frac{1}{2} \end{array}\right) \xrightarrow[(2/5)R_3\to R_3]{(1/2)R_2\to R_2} \left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 0 & 1 & \frac{3}{2} & \frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{5} & -\frac{1}{5} \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 0 & 1 & 3/2 & 3/2 & -1/2 \\ 0 & 0 & 1 & 1/5 & -1/5 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 0 & 1 & 3/2 & 3/2 & -1/2 \\ 0 & 0 & 1 & 1/5 & -1/5 \end{array}\right)$$

(Eliminate x_3 in Row 1 and Row 2 by Row 3)

$$\xrightarrow{-(3/2)R_3+R_2\to R_2} \left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 0 & 1 & 0 & 6/5 & -1/5 \\ 0 & 0 & 1 & 1/5 & -1/5 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 0 & 1 & 3/2 & 3/2 & -1/2 \\ 0 & 0 & 1 & 1/5 & -1/5 \end{array}\right)$$

(Eliminate x_3 in Row 1 and Row 2 by Row 3)

$$\xrightarrow{-(3/2)R_3+R_2\to R_2} \left(\begin{array}{cccc|c} 1 & -1 & 0 & -2 & 2 \\ 0 & 1 & 0 & 6/5 & -1/5 \\ 0 & 0 & 1 & 1/5 & -1/5 \end{array}\right)$$

$x = __ w$. $y = __ w$

(Eliminate x_2 in Row 1 by Row 2)

$$\xrightarrow{R_2+R_1\to R_1} \left(\begin{array}{cccc|c} 1 & 0 & 0 & -4/5 & 9/5 \\ 0 & 1 & 0 & 6/5 & -1/5 \\ 0 & 0 & 1 & 1/5 & -1/5 \end{array}\right)$$

$z = __ w$

x_4 is the free variable. The solutions set are of the form $(9/5 + (4/5)\alpha, -1/5 - (6/5)\alpha, -1/5 - (1/5)\alpha, \alpha)$, where α is a real number.

when $n = 2$

For a linear system of m equations and n unknowns, the number of solutions can be summarized as follows.

	$m > n$	$m = n$	$m < n$
no solution	most likely	possible	little chance
unique solution	little chance	most likely	impossible
infinity many solutions	little chance	possible	most likely

无解 唯一解 无数解

和重合一样 无数解

重合

$n = 2,\ m = 3 \Rightarrow$ solutions not (大概率)

For a linear system of m equations and n unknowns, the number of solutions can be summarized as follows.

	$m > n$	$m = n$	$m < n$
no solution	most likely	possible	little chance
unique solution	little chance	most likely	impossible
infinity many solutions	little chance	possible	most likely

无数解(多解)

Definition (Overdetermined System) A system of m linear equations in n unknowns is *overdetermined* if there are more equations than unknowns ($m > n$).

Definition (Underdetermined System) A system of m linear equations in n unknowns is *underdetermined* if there are fewer equations than unknowns ($m < n$).

Definition (Homogeneous Systems) A system of linear equations is said to be *homogeneous* if the constants on the righthand side are all zero.

Example

This system

$$\begin{cases} x - y - 2w = 0 \\ x + y + 3z + w = 0 \\ -y + z - w = 0 \end{cases}$$

is homogeneous.

Definition (Homogeneous Systems) A system of linear equations is said to be *homogeneous* if the constants on the righthand side are all zero.

Example

This system

$$\begin{cases} x - y - 2w = 0 \\ x + y + 3z + w = 0 \\ - y + z - w = 0 \end{cases}$$

is homogeneous.

Property of homogeneous systems Any homogeneous system is consistent because $(0, 0, \cdots, 0)$ is a solution of it. This solution $(0, 0, \cdots, 0)$ is said to be trivial.

Existence and Uniqueness

For a linear system $A\mathbf{x} = \mathbf{b}$, its solution(s) can be determined by answering the following questions in turn

1. Is the system consistent? (i.e. Does a solution **exist**?)
2. If a solution exists, is it **unique**?
3. What is the solution?

Existence and Uniqueness

For a linear system $A\mathbf{x} = \mathbf{b}$, its solution(s) can be determined by answering the following questions in turn

1. Is the system consistent? (i.e. Does a solution **exist**?)
2. If a solution exists, is it **unique**?
3. What is the solution?

Each question can be answered by the ref/rref of the augmented matrix $(A|\mathbf{b})$.

No Solution/Inconsistent

If the **row echelon form of the augmented matrix** contains a row of the form

$$\begin{pmatrix} 0 & \cdots & 0 & | & k \end{pmatrix}$$

for $k \neq 0$, the system is **inconsistent**. Otherwise, it is consistent.

No Solution/Inconsistent

If the **row echelon form of the augmented matrix** contains a row of the form

$$\begin{pmatrix} 0 & \cdots & 0 & | & k \end{pmatrix}$$

for $k \neq 0$, the system is **inconsistent**. Otherwise, it is consistent.

Example Consider

$$\begin{cases} x_1 & +2x_2 & -2x_3 & = 5 \\ & x_2 & -x_3 & = 2 \\ & x_2 & -x_3 & = 5 \end{cases}$$

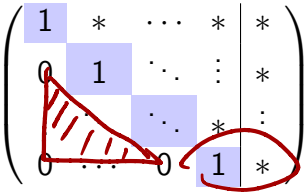
Then its augmented matrix

$$(A|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 2 & -2 & 5 \\ 0 & 1 & -1 & 2 \\ 0 & 1 & -1 & 5 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 3 \end{array}\right)$$

$0x_3 = -3$ is never true! The original system is inconsistent!

Unique

If the system is consistent and the nonzero rows of the **row echelon form** of the matrix form a strictly triangular system



the system will have a **unique solution** (**Consistent + No free variables**).

Infinitely Many

The system has **infinitely many solutions** if it is (**Consistent + At least 1 free variable**).

多解情况

Extra exercises

(a). Determine if the following matrices are in ref? rref? or neither?

row echelon form.

reduced

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right), \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right), \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{array}\right)$$

rref rref rref

$$\left(\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 4 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 1 & 5 \end{array}\right), \left(\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 4 \\ 0 & 1 & 3 & 0 & 1 \end{array}\right), \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{array}\right)$$

ref ref rref

Extra exercises

(b). Overdetermined? Underdetermined? Neither?

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right), \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right), \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{array}\right)$$

$m = 3,\ n = 2$

square $\Rightarrow$ neither

over determined

$m = 3$, $n = 4$

$$\left(\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 4 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 1 & 5 \end{array}\right), \left(\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 4 \\ 0 & 1 & 3 & 0 & 1 \end{array}\right), \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{array}\right)$$

under

over

Extra exercises

(c). Based on the following augmented matrices, determine if the corresponding linear systems are consistent or not.

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right), \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right), \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{array}\right)$$

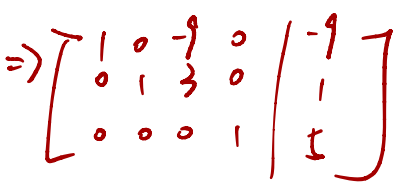
$$\left(\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 4 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 1 & 5 \end{array}\right), \left(\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 4 \\ 0 & 1 & 3 & 0 & 1 \end{array}\right), \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{array}\right)$$

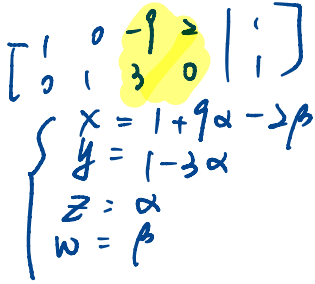
Extra exercises

(d). If consistent, how many solutions?

$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right), \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right), \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{array}\right)$$

$$\left(\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 4 \\ 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 1 & 5 \end{array}\right), \left(\begin{array}{cccc|c} 1 & 3 & 0 & 2 & 4 \\ 0 & 1 & 3 & 0 & 1 \end{array}\right), \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{array}\right)$$





Chapter 1 Matrices and System of Equations

Section 1.5 Elementary Matrices

Definition (Elementary Matrix)

Perform exactly one elementary row operation on the identity matrix I, the resulting matrix is called an **elementary matrix**.

Three types of elementary matrices corresponding to the three types of elementary row operations.

(I) $R_i \leftrightarrow R_j$

(II) $cR_i \to R_i, c \neq 0$

(III) $cR_j + R_i \to R_i$

Type I elementary matrix:
A matrix obtained by interchanging two rows of I.

Example Exchanging the ith row and j^{th} row of I_n, we obtain a Type I elementary matrix

$$E = \begin{bmatrix} I_{i-1} & & & \\ & 0 & 1 & \\ & & I_{j-i-1} & \\ & 1 & 0 & \\ & & & I_{n-j} \end{bmatrix}_{n \times n}$$

Type II elementary matrix:
A matrix obtained by multiplying a row of I by a nonzero constant, α say.

Example Multiplying the ith row of I_n by a nonzero real number α, we have a type II elementary matrix:

$$E = \text{diag}(1, \ldots, 1, \alpha, 1, \ldots, 1)$$

Type III elementary matrix:
A matrix obtained from I by adding a multiple of one row to another row.

Example Replacing R_i of A by $\alpha R_j + R_i$, we have a Type III elementary matrix

$$E = \begin{matrix} \\ i \\ \\ \\ j \\ \\ \end{matrix} \begin{bmatrix} I_{i-1} & & & & \\ & 1 & & \alpha & \\ & & I_{j-i-1} & & \\ & & & 1 & \\ & & & & I_{n-j} \end{bmatrix}$$

Theorem Let E be an elementary matrix of size $n \times n$.

1. For any $m \times n$ matrix A, EA is the matrix obtained when the same row operation is performed on A.
2. For any $n \times r$ matrix B, BE is the matrix obtained when the same column operation is performed on B.

Example (Type I)

exchange row 1 & 2

$$EA = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \end{pmatrix} = \begin{pmatrix} a_{21} & a_{22} & \cdots & a_{2n} \\ a_{11} & a_{12} & \cdots & a_{1n} \\ a_{31} & a_{32} & \cdots & a_{3n} \end{pmatrix}$$

exchange row 1 & 2

left

$$BE = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{pmatrix} b_{12} & b_{11} & b_{13} \\ b_{22} & b_{21} & b_{23} \\ b_{32} & b_{31} & b_{33} \end{pmatrix}.$$

right $\Rightarrow$ thinking about the column.

Example (Type II)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ \alpha a_{21} & \alpha a_{22} & \alpha a_{23} & \alpha a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix}$$

Example (Type III)

the operations to E when multiplied with A would do the same to A and then get the result

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} - 2a_{31} & a_{22} - 2a_{32} & a_{23} - 2a_{33} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

Left multiply E onto A = The same Elementary Row Operation on A.

Right multiply E onto A = The same Elementary Column Operation on A.

Extra exercises

Given that $A = \begin{pmatrix} 1 & -2 & 0 \\ 2 & 0 & 1 \\ 0 & -3 & 1 \end{pmatrix}$, and elementary matrices

$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$, $E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \beta \end{pmatrix}$, $E_3 = \begin{pmatrix} 1 & 0 & \beta \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Find

1. E_1A and AE_1
2. E_2A and AE_2.
3. E_1E_3A and E_3E_2A.

基本矩阵

Theorem If E is an elementary matrix, then E is nonsingular and E^{-1} is an elementary matrix of the same type.

E^{-1}

Proof The inverse of an elementary matrix is constructed by doing the reverse row operation on I. E^{-1} will be obtained by performing the row operation which would carry E back to I.

If E is obtained by switching rows i and j, then E^{-1} is also obtained by switching rows i and j.

If E is obtained by multiplying row i by the scalar α, then E^{-1} is obtained by multiplying row i by the scalar $1/\alpha$.

If E is obtained by adding α times row i to row j, then E^{-1} is obtained by adding $-\alpha$ times row i from row j.

① 加法变减法 ②乘法变除法 ③交换则不变

Extra exercises

Find the inverse matrices for the following elementary matrices:

$$E_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{pmatrix}, E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Answer:

$$E_1^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, E_2^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0.2 \end{pmatrix}, E_3^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

How about the inverse of A?

$$A = E_1 E_2 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

$$A^{-1} = (E_1 E_2)^{-1} = E_2^{-1} E_1^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 0.2 \end{pmatrix}$$

Definition (Row equivalent) A matrix B is **row equivalent** to a matrix A if there exists a finite sequence $E_1, E_2, \ldots, E_k$ of elementary matrices such that

$$B = E_k E_{k-1} \cdots E_1 A.$$

left

In other words, B is row equivalent to A if B can be obtained from A by a finite number of row operations.

In particular, if two augmented matrices $(A|\mathbf{b})$ and $(U|\mathbf{c})$ are row equivalent, then $A\mathbf{x} = \mathbf{b}$ and $U\mathbf{x} = \mathbf{c}$ are equivalent systems.

Property of row equivalent matrices

I. If A is row equivalent to B, then B is row equivalent to A.

II. If A is row equivalent to B, and B is row equivalent to C, then A is row equivalent to C.

有传递性

Proof Exercise

proof: $A = (E_k \cdots E_1)^{-1} B = E_1^{-1} \cdots E_k^{-1} B$

Theorem (Equivalent Conditions for Nonsingularity)

Let A be an $n \times n$ matrix. The following are equivalent:

(a) A is nonsingular;

Theorem (Equivalent Conditions for Nonsingularity)

Let A be an $n \times n$ matrix. The following are equivalent:

(a) A is nonsingular;

(b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{0}$;

Theorem (Equivalent Conditions for Nonsingularity)

Let A be an $n \times n$ matrix. The following are equivalent:

(a) A is nonsingular;

(b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{0}$;

(c) A is row equivalent to I.

(Or simply, $\text{rref}(A) = I$. A can be written as a product of elementary matrices.)

(d) .....more in Chap 2-6

A is nonsingular $\rightarrow$ $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{0}$

Assume: 可逆 $A \cdot A^{-1} = I = A^{-1} \cdot A$

Proof $(a) \Rightarrow (b)$ If A is nonsingular and $\mathbf{x}_0$ is a solution of $A\mathbf{x} = \mathbf{0}$, then

$$\mathbf{x}_0 = I\mathbf{x}_0 = (A^{-1}A)\mathbf{x}_0 = A^{-1}(A\mathbf{x}_0) = A^{-1}\mathbf{0} = \mathbf{0}.$$

Thus, $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

$Ax = 0 \Rightarrow A^{-1} \cdot Ax = A^{-1}0 \Rightarrow x = 0$

(a) $\Rightarrow$ (b)

(c) $\Leftarrow$ (b)

$A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\rightarrow$ $\text{rref}(A) = I$



Proof $(b) \Rightarrow (c)$ If we use elementary row operations, the system can be transformed into the form $U\mathbf{x} = \mathbf{0}$, where U is in row echelon form. If one of the diagonal elements of U were 0, the last row of U would consist entirely of 0's. But then $A\mathbf{x} = \mathbf{0}$ would be equivalent to a system with more unknowns than equations and, hence, there would have a nontrivial solution. Thus, U must be a strictly triangular matrix with diagonal elements all equal to 1. Hence, I is the reduced row echelon form of A and A is row equivalent to I.

assume $Ax = 0 \Rightarrow x = 0$

proof by contradiction: If $\text{rref}(A) \neq I$. then we must have

$\Rightarrow$ infinity many solutions

$\text{rref}(A) = I \quad \rightarrow \quad A$ is nonsingular

Proof $(c) \Rightarrow (a)$ If A is row equivalent to I_n, or we can say I_n is row equivalent to A. Then there exist elementary matrices $E_1, E_2, \cdots, E_k$ such that

$$I_n = E_k E_{k-1} \cdots E_1 A$$

Since the inverse of a matrix is unique if it exists, then A is nonsingular and

$$A^{-1} = E_k E_{k-1} \cdots E_1.$$

proof: assume rref(A) = I (to proof A is nonsingular $\Rightarrow A^{-1}$ exist)
then $I = E_k \cdots E_1 A$ for some elementary matrices
$\Rightarrow A^{-1} = E_k \cdots E_1 \Rightarrow A$ is nonsingular

Corollary The system $A\mathbf{x} = \mathbf{b}$ of n linear equations in n unknowns has a unique solution **if and only if** A is nonsingular.

Proof If A is nonsingular and $\mathbf{x}_0$ is any solution of $A\mathbf{x} = \mathbf{b}$, then $A\mathbf{x}_0 = \mathbf{b}$. Multiplying both sides of this equation by A^{-1}, we must have $\mathbf{x}_0 = A^{-1}\mathbf{b}$.

Conversely, if $A\mathbf{x} = \mathbf{b}$ has a unique solution $\mathbf{x}_0$, then we claim that A cannot be singular. Indeed, if A were singular, then the equation $A\mathbf{x} = \mathbf{0}$ would have a solution $\mathbf{z} \neq \mathbf{0}$. But this would imply that $\mathbf{y} = \mathbf{x}_0 + \mathbf{z}$ is a second solution of $A\mathbf{x} = \mathbf{b}$, since

$$A\mathbf{y} = A(\mathbf{x}_0 + \mathbf{z}) = A\mathbf{x}_0 + A\mathbf{z} = \mathbf{b} + \mathbf{0} = \mathbf{b}$$

Therefore, if $A\mathbf{x} = \mathbf{b}$ has a unique solution, then A must be nonsingular.

Suppose A is singular. Let b be fixed

① no sol

② a solution x_0 i.e. $Ax_0 = b$

$x^* \neq 0$ s.t. $Ax^* = 0$

$\Rightarrow$ $x_0 + x^*$ is also a sol $\Rightarrow$ $A(x^* + x_0) = 0 + b = b$

$\Rightarrow$ not unique

In fact. $x_0 + \alpha x^*$ is a solution for many $\alpha \in \mathbb{R}$

Steps to compute the inverse of an $n \times n$ matrix A

1. Form the $n \times 2n$ matrix $[A|I_n]$
2. Compute rref$(A|In)$
3. If rref$(A|I_n) = (I_n|C)$, then $A^{-1} = C$. Otherwise, A is singular.

Why it work? If A is nonsingular, then A is row equivalent to I and hence there exist elementary matrices $E_1, \cdots, E_k$ such that

$$\begin{aligned}(A|I_n) &\to E_1(A|I_n) = (E_1A|E_1)\\ &\to E_2(E_1A|E_1) = (E_2E_1A|E_2E_1)\\ &\vdots\\ &\to E_k(E_{k-1}\cdots E_1A|E_{k-1}\cdots E_1) = (E_kE_{k-1}\cdots E_1A|E_kE_{k-1}\cdots E_1)\end{aligned}$$

If $(E_kE_{k-1}\cdots E_1A|E_kE_{k-1}\cdots E_1) = (I_n|C)$, then $E_kE_{k-1}\cdots E_1A = I_n$ and $E_kE_{k-1}\cdots E_1 = C$, giving $A^{-1} = E_kE_{k-1}\cdots E_1 = C$.

Example Let $A = \begin{pmatrix} 1 & 3 & 1 \\ 2 & 0 & -1 \\ 1 & 2 & 0 \end{pmatrix}$. Find inverse of A.

Solution

Identity matrix

$$\left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 2 & 0 & -1 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right) \xrightarrow[-R_1+R_3\to R_3]{-2R_1+R_2\to R_2} \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & -6 & -3 & -2 & 1 & 0 \\ 0 & -1 & -1 & -1 & 0 & 1 \end{array}\right)$$

A $\quad$ I_n

$$\xrightarrow{-1/6R_2+R_3\to R_3} \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & -6 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1/2 & -2/3 & -1/6 & 1 \end{array}\right)$$

make $A \to \text{rref} \to I_n$

$I_n \to$ same operation $\to A^{-1}$

$$\xrightarrow[-2R_3 \to R_3]{-1/6R_2 \to R_2} \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1/2 & 1/3 & -1/6 & 0 \\ 0 & 0 & 1 & 4/3 & 1/3 & -2 \end{array}\right) \xrightarrow[-(1/2)R_3+R_2 \to R_2]{-R_3+R_1 \to R_1}$$

$$\left(\begin{array}{ccc|ccc} 1 & 3 & 0 & -1/3 & -1/3 & 2 \\ 0 & 1 & 0 & -1/3 & -1/3 & 1 \\ 0 & 0 & 1 & 4/3 & 1/3 & -2 \end{array}\right) \xrightarrow{-3R_2+R_1 \to R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 2/3 & 2/3 & -1 \\ 0 & 1 & 0 & -1/3 & -1/3 & 1 \\ 0 & 0 & 1 & 4/3 & 1/3 & -2 \end{array}\right)$$

$$A^{-1} = \begin{pmatrix} 2/3 & 2/3 & -1 \\ -1/3 & -1/3 & 1 \\ 4/3 & 1/3 & -2 \end{pmatrix} \to A^{-1}$$

$\Rightarrow$ check $A \cdot A^{-1} = I$

Extra Example on matrix inverse*

Find the inverse matrix of A, where

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$$

.

Answer:

$$A^{-1} = \begin{bmatrix} -0.1 & 0.3 \\ 0.4 & -0.2 \end{bmatrix}$$

Definition (Triangular matrices)

1. A square matrix $U = (u_{ij})$ is *upper triangular* if $u_{ij} = 0$ for $i > j$.
2. A square matrix $L = (l_{ij})$ is *lower triangular* if $l_{ij} = 0$ for $i < j$.
3. A matrix is *triangular* if it is either upper triangular or lower triangular.
4. A matrix is *unit lower (upper respectively) triangular* if it is a lower (upper respectively) triangular matrix with 1's on the diagonal.

Example

$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 5 \\ 0 & 0 & 6 \end{pmatrix}$ is upper triangular. $\begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}$ is lower triangular.

$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix}$ is unit upper triangular. $\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 5 & 1 \end{pmatrix}$ is unit lower triangular.

分解因式

Definition (LU factorization) The factorization of the matrix A into a product of a unit lower triangular matrix L times an upper triangular matrix U, i.e.,

$$A_{n\times n} = LU = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ \star & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ \star & \cdots & \star & 1 \end{bmatrix} \begin{bmatrix} \blacktriangle & \blacktriangle & \cdots & \blacktriangle \\ 0 & \blacktriangle & \ddots & \vdots \\ \vdots & \ddots & \ddots & \blacktriangle \\ 0 & \cdots & 0 & \blacktriangle \end{bmatrix}.$$

$(E_k \cdots E_1)A = U$

How to find L and U?

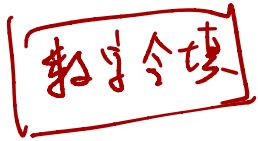
$A = (E_k \cdots E_1)^{-1} U \Rightarrow L$

A square matrix A is row equivalent to an upper triangular matrix U using only elementary matrix of type III that add a multiple of one row to another row **below** it. Thus, there exist a sequence of unit lower triangular elementary matrices $E_1, \cdots, E_k$ s.t.

$$E_k \cdots E_1 A = U,$$

and $L = (E_k \cdots E_1)^{-1} = E_1^{-1} \cdots E_k^{-1}$ is a unit lower triangular matrix.

operations



How to use LU factorization? Why do we need it?

LU factorization in solving a linear system:

The system $A\mathbf{x} = \mathbf{b}$ becomes $LU\mathbf{x} = \mathbf{b}$. Therefore, we have

$$L(U\mathbf{x}) = \mathbf{b}.$$

Let $\mathbf{y} = U\mathbf{x}$, we can find $\mathbf{x}$ by solving the following two systems of equations

$$L\mathbf{y} = \mathbf{b}, \quad U\mathbf{x} = \mathbf{y}.$$

Extra Example on LU Factorization*

Find the LU-decomposition of A, where

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$$

.

Answer:

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & -5 \end{bmatrix}.$$

Example

$$\begin{pmatrix} 1 & 3 & 1 \\ 2 & 0 & -1 \\ 1 & 2 & 0 \end{pmatrix} \xrightarrow[-1R_1+R_3\to R_3]{-2R_1+R_2\to R_2} \begin{pmatrix} 1 & 3 & 1 \\ 0 & -6 & -3 \\ 0 & -1 & -1 \end{pmatrix} \xrightarrow{\frac{-1}{6}R_2+R_3\to R_3} \begin{pmatrix} 1 & 3 & 1 \\ 0 & -6 & -3 \\ 0 & 0 & \frac{-1}{2} \end{pmatrix}$$

$$\text{Let } E_1 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{1}{6} & 1 \end{pmatrix}$$

Then

$$E_3E_2E_1A = U$$

Since $E_3E_2E_1A = U$, we have $A = E_1^{-1}E_2^{-1}E_3^{-1}U$.
Take $L = E_1^{-1}E_2^{-1}E_3^{-1}$.

$$
\begin{aligned}
L =& E_1^{-1}E_2^{-1}E_3^{-1} \\
=& \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{1}{6} & 1 \end{pmatrix}^{-1} \\
=& \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{1}{6} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & \frac{1}{6} & 1 \end{pmatrix}
\end{aligned}
$$

Remark This method does not work if A cannot be reduced into an upper triangular matrix using only type III row operation.

Extra Exercise*

Find the LU-decomposition of A, where

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 1 & 4 \\ 3 & 4 & 6 \end{bmatrix}$$

.

Answer:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1.5 & 0.1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 \\ 0 & -5 & 2 \\ 0 & 0 & 4.3 \end{bmatrix}$$

Remark This method does not work if A cannot be reduced into an upper triangular matrix using only *Type III* row operation by adding a multiple of one row to another row *below* it.

Example

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

.

Chapter 1 Matrices and System of Equations

More on Triangular Matrices

Properties on Triangular Matrices

1. The sum of two upper (lower) triangular matrices result in an upper (lower) triangular matrix.

$$\text{Upper} + \text{Upper} = \text{Upper}$$

2. The scalar multiple of an upper (lower) triangular matrix is an upper (lower) triangular matrix.

$$c\ \text{Upper} = \text{Upper}, \quad c \in \mathbf{R}.$$

3. The transpose of an upper (lower) triangular matrix is a lower (upper) triangular matrix.

$$\text{Upper}^T = \text{Lower}$$

transposes

Properties on Triangular Matrices

4. The multiplication of two upper (lower) triangular matrices is an upper (lower) triangular matrix.

$$\text{Upper Upper} = \text{Upper}$$

5. The inverse of an upper (lower) triangular matrix is an upper (lower) triangular matrix, given it is invertible.

$$\text{Upper}^{-1} = \text{Upper}, \quad \text{given nonsingular.}$$

6. The kth power of an upper (lower) triangular matrix is an upper (lower) triangular matrix.

$$\text{Upper}^k = \text{Upper}, \quad k \text{ is a positive integer.}$$

Specially for diagonal matrices, above properties hold and the resulting matrices stay as diagonal.

Example Find the result for the following matrices after power 3:

Diagonal matrix $D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ and lower triangular $L = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$.

Based on your observation, take a guess on

$$D^k_{n\times n} = \begin{pmatrix} d_1 & 0 & \cdots & 0 & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & d_{n-1} & 0 \\ 0 & 0 & \cdots & 0 & d_n \end{pmatrix}^k = ?$$

Example Find DL and DA:

$$D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}, L = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad \text{and} \quad A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Answers:

$$DL = \begin{pmatrix} 3 & 0 & 0 \\ -2 & 2 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad \text{and} \quad DA = \begin{pmatrix} 3 & 3 & 3 \\ 2 & 2 & 2 \\ 0 & -1 & -1 \end{pmatrix}.$$

Based on your observation, take a guess on

$$\begin{pmatrix} d_1 & 0 & \cdots & 0 & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & d_{n-1} & 0 \\ 0 & 0 & \cdots & 0 & d_n \end{pmatrix} A_{n\times k} = ?$$

Chapter 1 Matrices and System of Equations

Section 1.6 Partitioned Matrices

Definition (Blocks) A matrix can be partitioned into smaller matrices by drawing horizontal lines between the rows and vertical lines between the columns. The smaller matrices are often referred to as *blocks*.

Example

$$A = \left(\begin{array}{ccc|cc} 1 & 2 & 3 & 4 & 5 \\ -2 & 4 & -7 & 9 & 3 \\ \hline 7 & 6 & 5 & 9 & 8 \end{array}\right) = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

where $A_{11} = \begin{pmatrix} 1 & 2 & 3 \\ -2 & 4 & -7 \end{pmatrix}$, $A_{12} = \begin{pmatrix} 4 & 5 \\ 9 & 3 \end{pmatrix}$

$$A_{21} = \begin{pmatrix} 7 & 6 & 5 \end{pmatrix}, A_{22} = \begin{pmatrix} 9 & 8 \end{pmatrix}$$

Example

$$A = \left(\begin{array}{c|c|c|c|c} 1 & 2 & 3 & 4 & 5 \\ -2 & 4 & -7 & 9 & 3 \\ 7 & 6 & 5 & 9 & 8 \end{array}\right) = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4 & \mathbf{a}_5 \end{pmatrix}$$

where $\mathbf{a}_1 = \begin{pmatrix} 1 \\ -2 \\ 7 \end{pmatrix}$, $\mathbf{a}_2 = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$, $\mathbf{a}_3 = \begin{pmatrix} 3 \\ -7 \\ 5 \end{pmatrix}$, $\mathbf{a}_4 = \begin{pmatrix} 4 \\ 9 \\ 9 \end{pmatrix}$, $\mathbf{a}_5 = \begin{pmatrix} 5 \\ 3 \\ 8 \end{pmatrix}$.

Example

$$B = \left(\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ \hline -2 & 4 & -7 & 9 & 3 \\ \hline 7 & 6 & 5 & 9 & 8 \end{array}\right) = \begin{pmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \end{pmatrix}$$

where $\mathbf{b}_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \end{pmatrix}$, $\mathbf{b}_2 = \begin{pmatrix} -2 & 4 & -7 & 9 & 3 \end{pmatrix}$,
$\mathbf{b}_3 = \begin{pmatrix} 7 & 6 & 5 & 9 & 8 \end{pmatrix}$,

Block matrix multiplication If the blocks have the proper dimensions, the block multiplication can be carried out in the same manner as ordinary matrix multiplication.

For example, for $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ and $B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$,

$$AB = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}$$

Example

$$A=\left(\begin{array}{ccc|cc} 1 & 2 & 3 & 4 & 5 \\ -2 & 4 & -7 & 9 & 3 \\ \hline 7 & 6 & 5 & 9 & 8 \end{array}\right)=\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},\quad B=\left(\begin{array}{cc} 3 & 5 \\ 2 & 6 \\ 1 & 7 \\ \hline -1 & -2 \\ -3 & -4 \end{array}\right)=\begin{pmatrix} B_1 \\ B_2 \end{pmatrix}.$$

$$AB=\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}\begin{pmatrix} B_1 \\ B_2 \end{pmatrix}=\begin{pmatrix} A_{11}B_1+A_{12}B_2 \\ A_{21}B_1+A_{22}B_2 \end{pmatrix}$$

block wise

size match

$$=\begin{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ -2 & 4 & -7 \end{pmatrix}\left(\begin{array}{cc} 3 & 5 \\ 2 & 6 \\ 1 & 7 \end{array}\right)+\begin{pmatrix} 4 & 5 \\ 9 & 3 \end{pmatrix}\begin{pmatrix} -1 & -2 \\ -3 & -4 \end{pmatrix} \\ \begin{pmatrix} 7 & 6 & 5 \end{pmatrix}\left(\begin{array}{cc} 3 & 5 \\ 2 & 6 \\ 1 & 7 \end{array}\right)+\begin{pmatrix} 9 & 8 \end{pmatrix}\begin{pmatrix} -1 & -2 \\ -3 & -4 \end{pmatrix} \end{pmatrix}=\begin{pmatrix} -9 & 10 \\ -23 & -65 \\ \hline 5 & 56 \end{pmatrix}$$

The product AB is the same no matter A or B is treated as partitioned matrices or not.

$$AB = \left(\begin{array}{ccc|cc} 1 & 2 & 3 & 4 & 5 \\ -2 & 4 & -7 & 9 & 3 \\ \hline 7 & 6 & 5 & 9 & 8 \end{array}\right) \left(\begin{array}{cc} 3 & 5 \\ 2 & 6 \\ 1 & 7 \\ \hline -1 & -2 \\ -3 & -4 \end{array}\right) = \left(\begin{array}{cc} -9 & 10 \\ -23 & -65 \\ \hline 5 & 56 \end{array}\right).$$

$$AB = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ -2 & 4 & -7 & 9 & 3 \\ 7 & 6 & 5 & 9 & 8 \end{pmatrix} \begin{pmatrix} 3 & 5 \\ 2 & 6 \\ 1 & 7 \\ -1 & -2 \\ -3 & -4 \end{pmatrix} = \begin{pmatrix} -9 & 10 \\ -23 & -65 \\ 5 & 56 \end{pmatrix}.$$

If the blocks have the proper dimensions, the block multiplication can be carried out in the same manner as ordinary matrix multiplication.

Example

Let $A = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ -2 & 4 & -7 & 9 & 3 \\ 7 & 6 & 5 & 9 & 8 \end{pmatrix}$, $B = \left(\begin{array}{c|c} 3 & 5 \\ 2 & 6 \\ 1 & 7 \\ -1 & -2 \\ -3 & -4 \end{array}\right) = \left(\begin{array}{c|c} \mathbf{b}_1 & \mathbf{b}_2 \end{array}\right)$. Since A is of size 3×5 and $\mathbf{b}_1$ is of size 5×1, the partition of B can be used to compute AB.

$$AB = \left(\begin{array}{cc} A\mathbf{b}_1 & A\mathbf{b}_2 \end{array}\right)$$

Example

Let

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ -2 & 4 & -7 & 9 & 3 \\ 7 & 6 & 5 & 9 & 8 \end{pmatrix}, B = \left(\begin{array}{cc} 3 & 5 \\ \hline 2 & 6 \\ \hline 1 & 7 \\ \hline -1 & -2 \\ \hline -3 & -4 \end{array}\right) = \begin{pmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \mathbf{b}_3 \\ \mathbf{b}_4 \\ \mathbf{b}_5 \end{pmatrix}.$$

Since A is of size 3×5 and $\mathbf{b}_1$ is of size 1×2, the partition of B cannot be used to compute AB.

Exercise Let $B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \left(\begin{array}{cc|cc} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ \hline 3 & 1 & 1 & 1 \\ 3 & 2 & 1 & 2 \end{array}\right)$ and $D = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$.

Find $\begin{pmatrix} D & O \\ O & I \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$.

Analysis before writing the solution

Since D has two rows, O has 2 rows.

Since B has 4 rows, $\begin{pmatrix} D & O \\ O & I \end{pmatrix}$ has to have 4 columns in order to have multiplication. So O has two columns and hence is of size 2×2.

Since O has two columns, I has 2 columns and hence is of size 2×2.

Since D has two columns, O has 2 columns.

Since I has 2 columns, O has 2 rows and hence of size 2×2.

Example Let $D = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \left(\begin{array}{cc|cc} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ \hline 3 & 1 & 1 & 1 \\ 3 & 2 & 1 & 2 \end{array}\right)$.

Find $\begin{pmatrix} D & O \\ O & I \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$.

Solution

$$\begin{pmatrix} D & O \\ O & I \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \begin{pmatrix} DB_{11} + OB_{21} & DB_{12} + OB_{22} \\ OB_{11} + IB_{21} & OB_{12} + IB_{22} \end{pmatrix}$$

$$= \begin{pmatrix} DB_{11} & DB_{12} \\ B_{21} & B_{22} \end{pmatrix}$$

$$= \left(\begin{array}{cc|cc} 2 & 2 & 2 & 2 \\ 2 & 4 & 2 & 2 \\ \hline 3 & 1 & 1 & 1 \\ 3 & 2 & 1 & 2 \end{array}\right)$$

Example

Suppose A and C are $m \times m$ matrices, B and D are $n \times n$ matrices. Then

$$\left(\begin{array}{c|c} A & O_{m\times n} \\ \hline O_{n\times m} & B \end{array}\right)\left(\begin{array}{c|c} C & O_{m\times n} \\ \hline O_{n\times m} & D \end{array}\right)=\left(\begin{array}{c|c} AC & O_{m\times n} \\ \hline O_{n\times m} & BD \end{array}\right).$$

In particular, for any positive integer k, we have

$$\left(\begin{array}{cc|ccc} \lambda & 1 & 0 & 0 & 0 \\ 0 & \lambda & 0 & 0 & 0 \\ \hline 0 & 0 & \lambda & 1 & 0 \\ 0 & 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & 0 & \lambda \end{array}\right)^k = \left(\begin{array}{cc|ccc} \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}^k & & 0 & 0 & 0 \\ & & 0 & 0 & 0 \\ \hline 0 & 0 & & & \\ 0 & 0 & & \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}^k & \\ 0 & 0 & & & \end{array}\right).$$

Theorem Let A be an $n \times n$ square matrix with $A = \begin{pmatrix} A_{11} & O \\ O & A_{22} \end{pmatrix}$, where A_{11} has a size of $k \times k$ and $k < n$. Then A is nonsingular if and only if A_{11} and A_{22} are nonsingular. In this case we have

$$A^{-1} = \begin{pmatrix} A_{11}^{-1} & O \\ O & A_{22}^{-1} \end{pmatrix}.$$

Proof of the "if" part If A is nonsingular, then let $B = A^{-1}$ and partition B in the same manner as A. Since $BA = I = AB$,

$$\begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} \begin{pmatrix} A_{11} & O \\ O & A_{22} \end{pmatrix} = \begin{pmatrix} I_k & O \\ O & I_{n-k} \end{pmatrix} = \begin{pmatrix} A_{11} & O \\ O & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$$

$$\begin{pmatrix} B_{11}A_{11} & B_{12}A_{22} \\ B_{21}A_{11} & B_{22}A_{22} \end{pmatrix} = \begin{pmatrix} I_k & O \\ O & I_{n-k} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11} & A_{11}B_{12} \\ A_{22}B_{21} & A_{22}B_{22} \end{pmatrix}$$

Namely,

$$B_{11}A_{11} = I_k = A_{11}B_{11}, \qquad B_{22}A_{22} = I_{n-k} = A_{22}B_{22}.$$

Hence, A_{11} and A_{22} are both nonsingular and $A_{11}^{-1} = B_{11}$, $A_{22}^{-1} = B_{22}$.